



Debashis Majumder

Roll No.: 38732523007

BTech in Artificial Intelligence

Alipurduar Government Engineering and Management

College

+91-7001095441

✉ dm2707@ai.agemc.ac.in

✉ debashisxmajumder@gmail.com

🐙 GitHub

🌐 Website 🔗 LinkedIn Profile

EDUCATION

- **Alipurduar Government Engineering and Management College** 2023-27
BTech in Artificial Intelligence Pursuing
- **Barajaguli Gopal Academy (WBCHSE)** 2022-23
Physics, Chemistry and Mathematics Percentage: 65.04
- **Barajaguli Gopal Academy (WBBSE)** 2020-21
Secondary Education Percentage: 77.7

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, C++, SQL

Frameworks: Flask, Pandas, NumPy, Matplotlib, Scikit-Learn, OpenCV, PyTorch

Core Areas: Machine Learning, Deep Learning, Computer Vision, Data Preprocessing

Other Skills: Git, SQL, Docker, Linux, MySQL

RELEVANT COURSEWORK

- Machine Learning
- Deep Learning
- Artificial Intelligence
- Data Structures & Algorithms
- Design & Analysis of Algorithms
- Probability & Statistics
- Database Management Systems
- Operating Systems
- Computer Networks

PROJECTS

- **Plant Disease Classification (Ongoing)** *Python, PyTorch, OpenCV*
Applying attention-driven multi-ViT architecture to plant disease detection.
 - Developing a multi-Vision Transformer pipeline that uses attention scores to weight informative regions of leaf images for fine-grained disease classification.
 - Designing a data preprocessing and augmentation pipeline for plant disease image datasets to improve model generalization.
 - Benchmarking attention-weighted multi-ViT performance against baseline CNN architectures to quantify accuracy gains.
- **Smart Attendance System** *Python, Flask, InsightFace, scikit-learn*
Two-factor biometric attendance system combining fingerprint and face recognition.
 - Designed a face-recognition pipeline using RetinaFace for detection and ArcFace (InsightFace Buffalo_L) embeddings, with a KNN classifier and multi-frame majority voting to verify student identity from live camera input.
 - Implemented a two-factor verification layer that cross-checks the fingerprint sensor's identified roll number against the face-recognition model's prediction, rejecting attendance on mismatch or low-confidence detections.
 - Contributed to Flask REST API endpoints handling ESP32-CAM image uploads and fingerprint events, integrating the recognition pipeline with a SQLAlchemy-backed attendance database.
- **Wireless Data Transfer (WDT)** *Python, Flask, Raspberry Pi*
Secure Flask-based wireless file transfer system with QR-code access and SSH-gated admin control.
 - Built a production-ready wireless file transfer system for local networks, deployed on Raspberry Pi Zero 2W.
 - Implemented general and token-based secure upload modes with QR-code access and 24-hour auto-expiry.
 - Designed an SSH-tunnel-only admin dashboard, eliminating exposed admin credentials over the network.
 - Added multithreaded background cleanup, session handling, and CSV-based audit logging for all file activity.
 - Supported cross-device access (mobile/desktop) with production deployment via Gunicorn and systemd.

CERTIFICATES

- **Certificate for Python for Data Science, AI and Development** from Coursera